

### Section 1: Diagnostic Criteria and Classification

Type 2 Diabetes Mellitus (T2D) is characterized by progressive loss of adequate beta-cell insulin secretion frequently on the background of insulin resistance.

Diagnostic criteria include: Fasting Plasma Glucose (FPG)  $\geq 126$  mg/dL (7.0 mmol/L) after an 8-hour fast; or 2-hour Post-Prandial Glucose  $\geq 200$  mg/dL (11.1 mmol/L) during an OGTT (75g); or Glycated Hemoglobin (HbA1c)  $\geq 6.5\%$  (48 mmol/mol); or random plasma glucose  $\geq 200$  mg/dL with classic hyperglycemic symptoms (polyuria, polydipsia, unexplained weight loss).

### Section 2: Initial Assessment and Lifestyle Intervention

All patients diagnosed with T2D must receive Medical Nutrition Therapy (MNT) and diabetes self-management education. Recommend at least 150 minutes of moderate-intensity aerobic physical activity per week, distributed over at least 3 days with no more than 2 consecutive days without exercise.

### Section 3: Pharmacological Management

**First-Line Pharmacotherapy:** Metformin remains the preferred initial pharmacological agent for the treatment of Type 2 diabetes unless contraindicated. Initial titration starts at 500 mg once or twice daily with meals to reduce gastrointestinal side effects, titrating to a target therapeutic dose of 2000 mg daily.

**Cardiorenal Risk Stratification:** In patients with established Atherosclerotic Cardiovascular Disease (ASCVD), high risk indicators, Heart Failure (HF), or Chronic Kidney Disease (CKD), SGLT2 inhibitors (e.g., Empagliflozin, Dapagliflozin) or GLP-1 receptor agonists (e.g., Semaglutide, Dulaglutide) with proven CVD/renal benefit are recommended independent of baseline HbA1c.

**Dual and Triple Therapy:** If glycemic targets are not achieved within 3 months of monotherapy, intensify treatment by adding an agent from a complementary class: SGLT2i, GLP-1 RA, DPP-4 inhibitors, or Pioglitazone.

### Section 4: Glycemic Targets and Hypoglycemia Protocols

**Glycemic Goals:** An HbA1c goal of < 7.0% (53 mmol/mol) is recommended for most non-pregnant adult patients. More stringent targets (< 6.5%) may be considered for younger patients with short disease duration and low hypoglycemia risk. Less stringent targets (< 7.5% - 8.0%) are appropriate for patients with limited life expectancy, extensive comorbidities, or severe hypoglycemia history.

**Hypoglycemia Management ('Rule of 15'):** When blood glucose drops below 70 mg/dL (3.9 mmol/L), administer 15-20 grams of fast-acting oral glucose (e.g., 4 glucose tablets, 1/2 cup fruit juice). Recheck blood glucose after 15 minutes. If still < 70 mg/dL, repeat treatment. Once normalized, provide a meal or complex carbohydrate snack.

## Section 5: Renal Considerations and Safety Thresholds

**Renal Function Monitoring:** Calculate eGFR at least annually in all diabetic patients. Metformin is contraindicated in patients with severe renal impairment (eGFR < 30 mL/min/1.73m<sup>2</sup>). In patients with eGFR between 30 and 44 mL/min/1.73m<sup>2</sup>, the maximum recommended Metformin dose is 1000 mg daily.

**SGLT2 Inhibitor Renal Limits:** Empagliflozin and Dapagliflozin may be initiated down to an eGFR of 20-25 mL/min/1.73m<sup>2</sup> for cardiorenal protection, though their glycemic efficacy decreases as eGFR declines.